

Peering Karmaşasına Son: AWS Transit Gateway ile Basit ve Güvenli Çoklu VPC Ağı

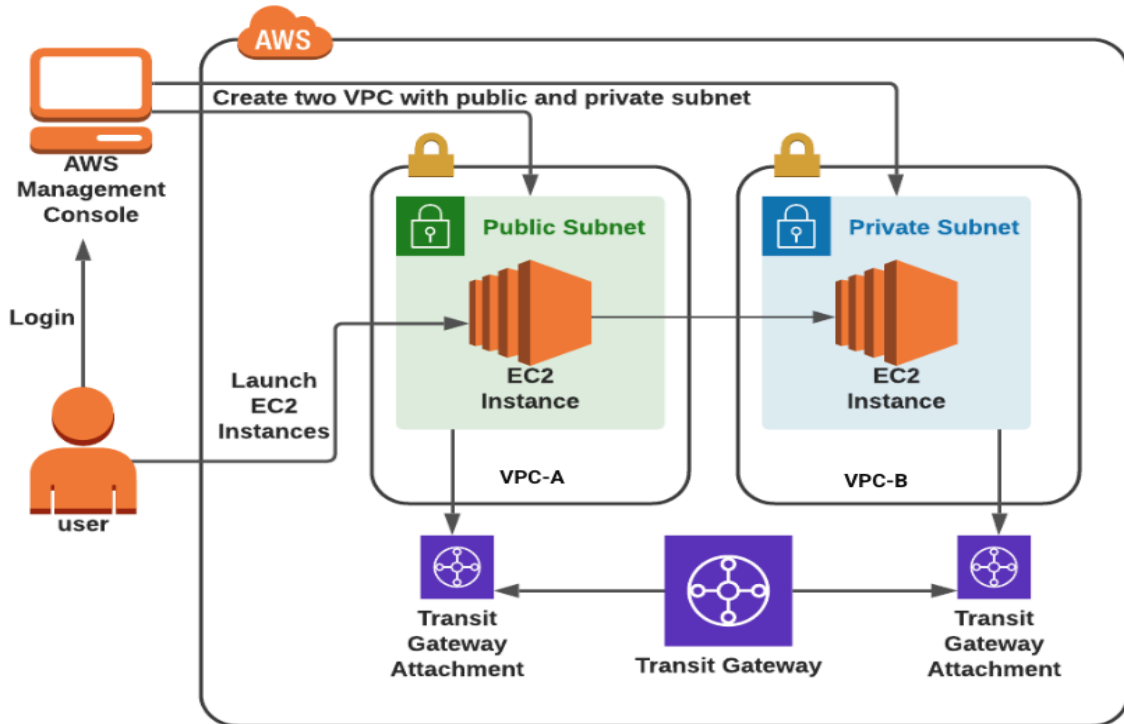
Hedef

- Bir **Transit Gateway (TGW)** oluşturmak.
- **VPC-A** ve **VPC-B** ağlarını Transit Gateway'e **attachment** ile bağlamak.
- **TGW Route Tables** ve **VPC Route Tables** yapılandırarak **transitive routing** sağlamak.
- **Private IP** adresleri üzerinden iki VPC arasındaki iletişimi doğrulamak.

Mimari

Senaryomuzda:

- **VPC-A** içinde bir *public subnet* bulunacak ve buraya internet erişimi olan bir EC2 instance kurulacak.
 - **VPC-B** içinde ise *private subnet* olacak ve yalnızca iç iletişim için kullanılan bir EC2 instance yer alacak.
- İki VPC, **Transit Gateway** üzerinden haberleşecek.



Bu yapı sayesinde, merkezi bir ağ omurgası üzerinden farklı VPC'ler arası güvenli iletişim kurulmuş olacak.



Lab Adımları

1. VPC-A Oluşturma

- AWS Management Console'da **N. Virginia (us-east-1)** bölgesini seçin.
- **VPC only** seçeneği ile **VPC-A** isimli bir VPC oluşturun.
- IPv4 CIDR: **10.0.0.0/24**
- DNS Resolution ve DNS Hostnames ayarlarını aktif edin.

2. VPC-A İçinde Public Subnet Oluşturma

- **10.0.0.0/24** CIDR bloklu **Public_subnet_VPC-A** isimli subnet'i oluşturun.

3. Internet Gateway (IGW) Ekleme

- IGW oluşturun ve **VPC-A**'ya attach edin.

4. Public Route Table Yapılandırma

- **PublicRT** isimli bir route table oluşturun.
- Public subnet ile ilişkilendirin.
- 0.0.0.0/0 trafiğini IGW'ye yönlendiren rota ekleyin.

5. VPC-A İçinde EC2 Başlatma

- Amazon Linux 2023 AMI, **t2.micro** instance tipiyle **VPC-A_EC2** isimli bir sunucu açın.
- Public IP atanmış olacak.
- User Data ile Apache Web Server kurup bir karşılama sayfası yayınlayın.

6. VPC-B Oluşturma

- **20.0.0.0/24** CIDR bloklu **VPC-B**'yi oluşturun.
- DNS Resolution ve DNS Hostnames ayarlarını aktif edin.

7. VPC-B İçinde Private Subnet Oluşturma

- **Private_subnet_VPC-B** isimli subnet oluşturun.
- **PrivateRT** route table ile ilişkilendirin (şimdilik internete çıkışı olmayacak).

8. VPC-B İçinde EC2 Başlatma

- Amazon Linux 2023 AMI, **t2.micro** instance tipiyle **VPC-B_EC2** isimli bir sunucu açın.
- Public IP **atanmayacak**; yalnızca private IP olacak.

9. Transit Gateway (TGW) Oluşturma

- **DemoTG** isimli Transit Gateway oluşturun.
- Birkaç dakika içinde *available* durumuna gelecektir.

10. TGW Attachment'ları Oluşturma

- **VPC-A_TGA** ve **VPC-B_TGA** attachment'larını oluşturun.

11. Route Table Yapılandırma

- **VPC-A** → **PublicRT** içine **20.0.0.0/24** → **DemoTG** rotası ekleyin.
- **VPC-B** → **PrivateRT** içine **10.0.0.0/24** → **DemoTG** rotası ekleyin.

12. Bağlantıyı Test Etme

- **VPC-A_EC2**'ye SSH ile bağlanın.
- Private anahtar dosyasını kopyalayın.
- Buradan **VPC-B_EC2**'nin private IP'sine SSH yapın.
Eğer bağlantı başarılı olursa Transit Gateway üzerinden iki VPC arasında iletişim sağlanmış demektir .

13. Temizlik (Cleanup)

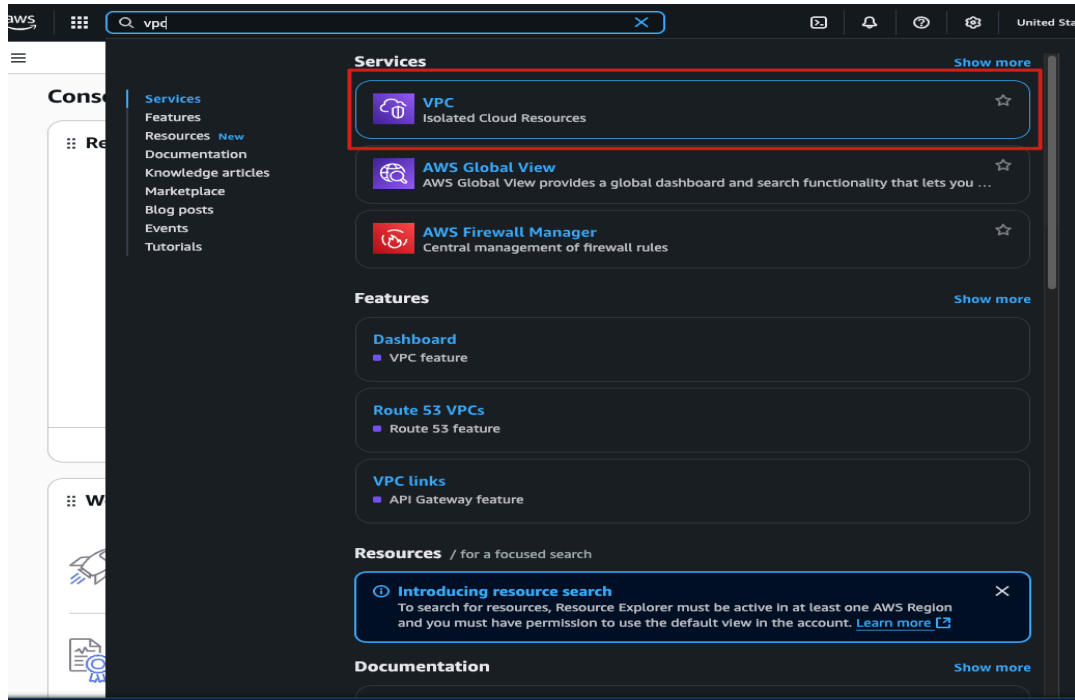
- Lab sonunda oluşturduğunuz tüm EC2, Subnet, Route Table, IGW ve Transit Gateway kaynaklarını silin.

AWS Transit Gateway ile Çoklu VPC Omurgası Kurulumu

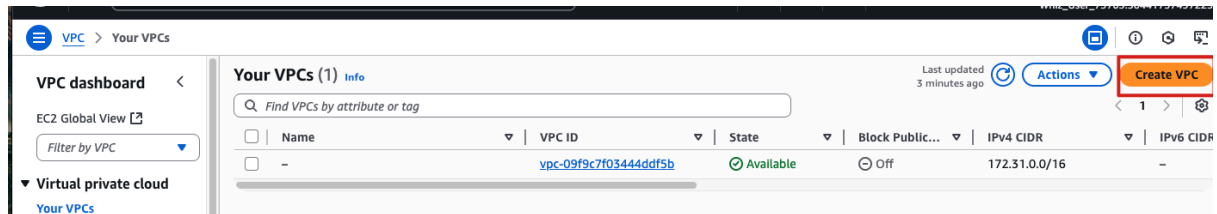
Bu bölümde, adım adım **VPC-A (public)** ve **VPC-B (private)** oluşturup bunları **Transit Gateway** üzerinden konuşturacağız.

Step 1: VPC-A oluşturma

İlk olarak internet erişimi olan **VPC-A**'yı oluşturacağız. AWS console ile login olduktan sonra **N. Virginia (us-east-1)** bölgesini seçerek servislerden VPC' yi seçin.



1. Sol menüden **Your VPCs** bulduktan sonra **Create VPC** buton ile VPC-A yı oluşturmaya başlayalım



- VPC only seç
- Name tag : VPC-A
- IPv4 CIDR : 10.0.0.0/24

☰ VPC > Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC-A

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
 10.0.0.0/24
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info
 Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q VPC-A X [Remove tag](#)

[Add tag](#)

You can add 49 more tags

[Cancel](#) [Preview code](#) [Create VPC](#)

2. Diğer ayarları default'ta bırakarak **Create VPC** butonuna tıklayın.
3. VPC başarıyla oluşturuldu. **VPC ID**'yi not alın

✓ You successfully created vpc-018efef110bc6f589 / VPC-A

vpc-018efef110bc6f589 / VPC-A Actions

Details Info

VPC ID vpc-018efef110bc6f589	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-063cb82d1929ed2f9	Main route table rtb-0fd99e7e2a74d147c
Main network ACL acl-01c76932903c99c98	Default VPC No	IPv4 CIDR 10.0.0.0/24	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 834266228480

4. Listedeki VPC-A'yı seçin → **Actions** → **Edit VPC settings**.

- **Enable DNS resolution** ve **Enable DNS hostnames** kutularını işaretleyip **Save**.

Edit VPC settings [Info](#)

VPC details

VPC ID
vpc-018efef110bc6f589

Name
VPC-A

DHCP settings

DHCP option set [Info](#)
dopt-063cb82d1929ed2f9

DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

Network Address Usage metrics settings

☐ Enable Network Address Usage metrics [Info](#)

[Cancel](#) [Save](#)

✓ Artık internet erişimi destekleyecek ilk VPC hazır.

Step 2: VPC-A içinde Public Subnet Oluşturma

Public subnet, dış dünyaya erişimi olan sunucular için kullanılır. Bu adımda internetten erişilebilecek bir **public subnet** oluşturacağız.

1. Sol menüden **Subnets** → **Create subnet**.

Subnets (6) [Info](#)

Find subnets by attribute or tag

Last updated 8 minutes ago [Actions](#) [Create subnet](#)

	Name	Subnet ID	State	VPC	Block Public...
☐	-	subnet-0bcf36eb14d543e7d	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-032c97cae236dc6af	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-09a87397ac2610fd8	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-0b16ab23e115d25d4	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-05b4c0af0d1763c3f	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-009b05dd22e8efdc8	Available	vpc-09f9c7f03444ddf5b	Off

Virtual private cloud

Your VPCs

[Subnets](#)

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

2. **VPC ID:** VPC-A
3. **Subnet name:** Public_subnet_VPC-A
4. **Availability Zone:** No Preference
5. **IPv4 subnet CIDR block:** 10.0.0.0/24
6. **Create subnet** butonuna tıklayın.

VPC > Subnets > Create subnet

VPC

VPC ID
Creates subnets in this VPC.
vpc-018efef110bc6f589 (VPC-A)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/24

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public_subnet_VPC-A
The name can be up to 255 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/24

IPv4 subnet CIDR block
10.0.0.0/24 256 IPs

▼ Tags - optional

Key	Value - optional
Name	Public_subnet_VPC-A

Add new tag
You can add 49 more tags.

Remove

Add new subnet

Cancel Create subnet

less than a minute ago

Find subnets by attribute or tag

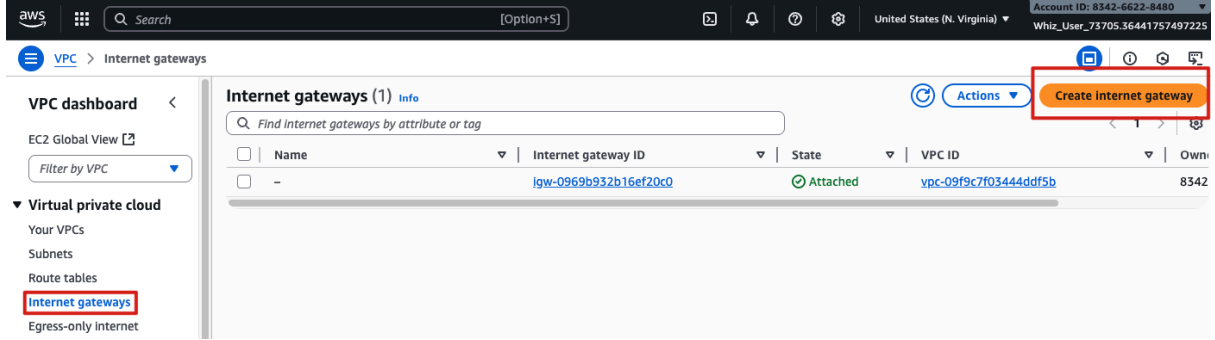
Subnet ID : subnet-03f93f5f64a529f75 Clear filters

	Name	Subnet ID	State	VPC	Block Public...
	Public_subnet_VPC-A	subnet-03f93f5f64a529f75	Available	vpc-018efef110bc6f589 VPC-A	Off

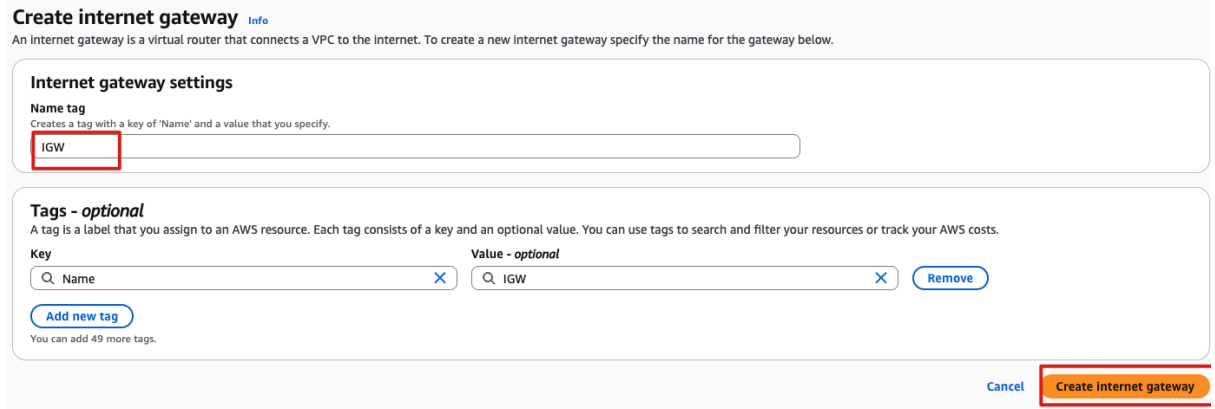
Step 3: Internet Gateway Oluştur ve VPC-A'ya Ekle

Public subnetteki EC2'nun internete çıkabilmesi için **Internet Gateway (IGW)** gerekli.

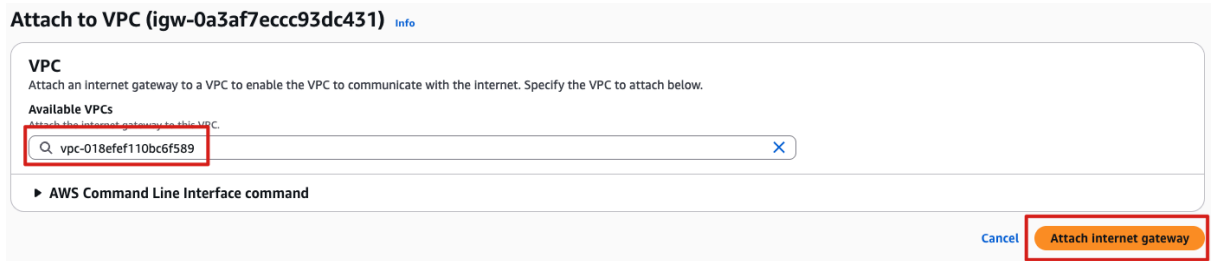
1. Sol menü: **Internet gateways** → **Create internet gateway**.



2. Name tag: **IGW** → **Create internet gateway**.



3. **Actions** → **Attach to VPC** → **Available VPCs: VPC-A** → **Attach**.



Internet gateway igw-0a3af7ecc93dc431 successfully attached to vpc-018efef110bc6f589

igw-0a3af7ecc93dc431 / IGW

[Actions](#)

Details [Info](#)

Internet gateway ID

igw-0a3af7ecc93dc431

State

Attached

VPC ID

vpc-018efef110bc6f589 | [VPC-A](#)

Owner

834266228480

Step 4: Public Route Table Oluştur ve Subnet ile İlişkilendir

Public subnet'in trafiğini IGW'ye yönetecek route table'ı oluşturacağız.

1. Sol panel: **Route tables** → **Create route table**.

The screenshot shows the AWS VPC console's 'Route tables' page. On the left, the 'Route tables' link is highlighted in the navigation menu. In the top right, the 'Create route table' button is highlighted with a red box. The main area displays a table of existing route tables:

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-08457cd051feca442	-	-	Yes	vpc-09f9c7fc
-	rtb-0fd99e7e2a74d147c	-	-	Yes	vpc-018efef1

2. Name: **PublicRT**
3. VPC: **VPC-A** → **Create route table**.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

PublicRT

VPC

The VPC to use for this route table

vpc-018efef110bc6f589 (VPC-A)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

PublicRT

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

4. Altteki **Subnet associations** sekmesine geç.
5. **Edit subnet associations** → **Public_subnet_VPC-A** seç → **Save associations**.

VPC > Route tables > rtb-0c50d333e4fec8fc0

VPC dashboard < EC2 Global View [?] Filter by VPC

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Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers New

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started Updated

rtb-0c50d333e4fec8fc0 / PublicRT

Actions

Details Info

Route table ID: rtb-0c50d333e4fec8fc0
Main: No
Explicit subnet associations: -
Edge associations: -
VPC: vpc-018efef110bc6f589 | VPC-A
Owner ID: 834266228480

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (0)

Find subnet association

Edit subnet associations

No subnet associations
You do not have any subnet associations.

Subnets without explicit associations (1)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public_subnet_VPC-A	subnet-03f93f5f64a529f75	10.0.0.0/24	-

VPC > Route tables > rtb-0c50d333e4fec8fc0 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public_subnet_VPC-A	subnet-03f93f5f64a529f75	10.0.0.0/24	-	Main (rtb-0fd99e7e2a74d147c)

Selected subnets

subnet-03f93f5f64a529f75 / Public_subnet_VPC-A

Cancel Save associations

Step 5: Public Route Ekle

Public subnet'in internete çıkışı için **0.0.0.0/0** → **IGW** rotası ekleyeceğiz.

1. **Route tables** → **PublicRT** seç.

VPC > Route tables > rtb-0c50d333e4fec8fc0

VPC dashboard < EC2 Global View [?] Filter by VPC

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Elastic IPs
Managed prefix lists
NAT gateways

rtb-0c50d333e4fec8fc0 / PublicRT

Actions

Details Info

Route table ID: rtb-0c50d333e4fec8fc0
Main: No
Explicit subnet associations: subnet-03f93f5f64a529f75 / Public_subnet_VPC-A
Edge associations: -
VPC: vpc-018efef110bc6f589 | VPC-A
Owner ID: 834266228480

Routes Subnet associations Edge associations Route propagation Tags

Routes (1)

Filter routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	Create Route Table

2. **Routes** sekmesi → **Edit routes** → **Add route**.
 - **Destination:** 0.0.0.0/0
 - **Target:** Internet Gateway → oluşturduğun IGW
3. **Save changes** ile kaydedin

Edit routes

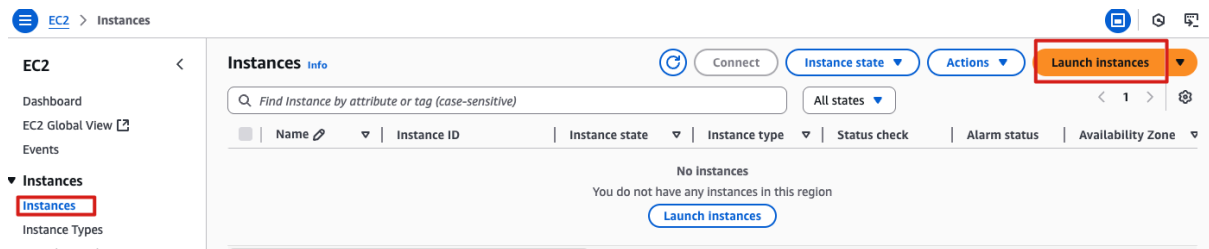
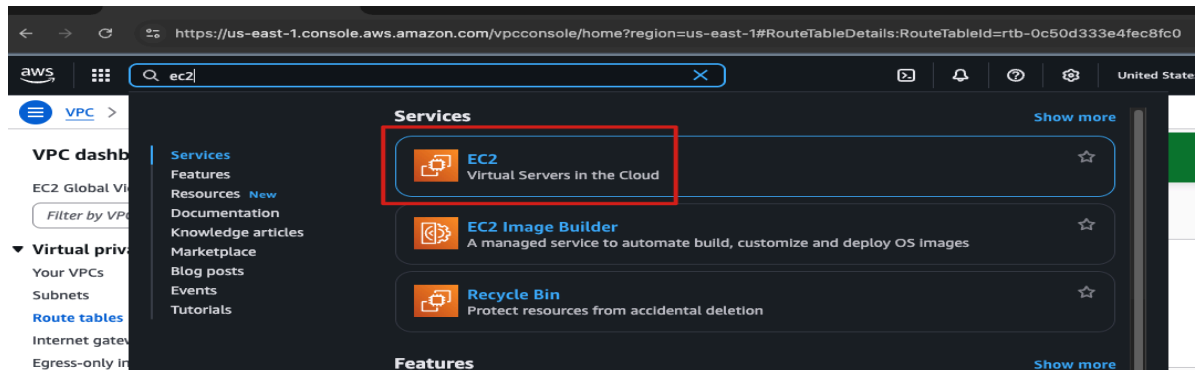
Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway		No	CreateRoute

Add route **Cancel** **Preview** **Save changes**

Step 6: VPC-A Public Subnet'te EC2 Başlat

Artık Public subnet içinde bir test sunucusu açalım.

1. Bölge: **us-east-1** (N. Virginia).
2. **Services** → **EC2** → **Instances** → **Launch instances**.



3. Name and tags:

- Name: **VPC-A_EC2**

4. Application and OS Images (AMI):

- Quick Start → Amazon Linux 2023 (kernel-6.1)

Name and tags Info

Name

VPC-A_EC2

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

My AMIs

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-00ca32bbc84273381 (64-bit (x86), uefi-preferred) / ami-0aa7db6294d00216f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

5. Instance type: **t2.micro**

▼ Instance type Info | Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible ▼

☐ All generations
[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

6. Key Pair (login):

- Create new key pair → Key pair name: **ec2_ssh_key**
- Type: RSA, File format: **.pem** → Create ve seç.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

ec2_ssh_key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

Key dosyasını bilgisayarına kaydet

7. Network settings → Edit:

- **VPC:** VPC-A
- **Subnet:** varsayılan (Public_subnet_VPC-A seçili gelmeli)
- **Auto-assign public IP:** Enable
- **Firewall (SG): Create a new security group**
 - **Security group name:** Public_EC2_SG
 - **Description:** Security group for public EC2
 - **Rules:**
 1. **SSH** → **Source:** Anywhere (0.0.0.0/0)
 2. **HTTP** → **Source:** Anywhere

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-018efef110bc6f589 (VPC-A)
10.0.0.0/24



Subnet [Info](#)

subnet-03f93f5f64a529f75 Public_subnet_VPC-A
VPC: vpc-018efef110bc6f589 Owner: 834266228480
Availability Zone: us-east-1d (use1-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.0.0/24



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

Public_EC2_SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/()#,@!+=&[]!*\$

Description - required [Info](#)

Security group for public EC2

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

[Remove](#)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

[Remove](#)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Custom

Source [Info](#)

Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

8. Advanced details:

- **User data:** aşağıdaki script'i yapıştır.

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul Public Server</h1><html>" >
/var/www/html/index.html
```

User data - optional | info

Upload a file with your user data or enter it in the field.

↑ Choose file

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul Server</h1><html>" >
/var/www/html/index.html
```

9. Launch instance. Instances sayfasında **running** olana kadar bekle.

Instances (1) [Info](#)Last updated less than a minute ago  [Connect](#) [Instance](#)

[All instances](#)

☐	Name 	Instance ID	Instance state	Instance type
☐	VPC-A_EC2	i-069dec671266f688e	 Pending  	t2.micro

10. EC2'nun IPv4 Public IP adresini not et.

i-0ec2dcb90085c6e81 (VPC-A_EC2)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary [Info](#)

Instance ID i-0ec2dcb90085c6e81	Public IPv4 address 3.236.83.74 open address	Private IPv4 addresses 10.0.0.15
IPv6 address -	Instance state Running	Public DNS ec2-3-236-83-74.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-0-15.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-0-15.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.236.83.74 [Public IP]	VPC ID vpc-0aa0ac3399337e18f (VPC-A)	

11. Bu IP'yi tarayıcıya yapıştırdığında karşılama sayfasını görmelisin.



Step 7: VPC-B'yi Oluřturma

řimdi ikinci VPC'yi (**VPC-B**) oluřturacađız. Bu VPC, yalnızca **private subnet** ięerecek ve dođrudan internete ıkıřı olmayacak.

1. **Services** → **VPC** → **Your VPCs** → **Create VPC** seęin
2. **VPC only**.
3. **Name tag**: **VPC-B**
4. **IPv4 CIDR block**: **20.0.0.0/24**
5. **Create VPC** butonuna tıklayın.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

VPC-B

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
20.0.0.0/24
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q VPC-B X [Remove tag](#)

[Add tag](#)

You can add 49 more tags

[Cancel](#) [Preview code](#) [Create VPC](#)

✓ Artık ikinci VPC hazır.

6. VPC ID'yi not edin.

vpc-097810e84793630a3 / VPC-B

Details Info			
VPC ID  vpc-097810e84793630a3	State ✔ Available	Block Public Access ⊖ Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-063cb82d1929ed2f9	Main route table rtb-0a40cb14ccc6f3f10
Main network ACL acl-0afb3fd0b89f0447e	Default VPC No	IPv4 CIDR 20.0.0.0/24	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID  834266228480

7. Listed **VPC-B** → **Actions** → **Edit VPC settings** →
Enable DNS resolution ve **Enable DNS hostnames** işaretle → **Save**.

VPC > Your VPCs > [vpc-097810e84793630a3](#) > Edit VPC settings

ⓘ ⌂ 🖨

Edit VPC settings [Info](#)

VPC details

VPC ID
 vpc-097810e84793630a3

Name
 VPC-B

DHCP settings

DHCP option set [Info](#)

dopt-063cb82d1929ed2f9

▼

DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

Network Address Usage metrics settings

☐ Enable Network Address Usage metrics [Info](#)

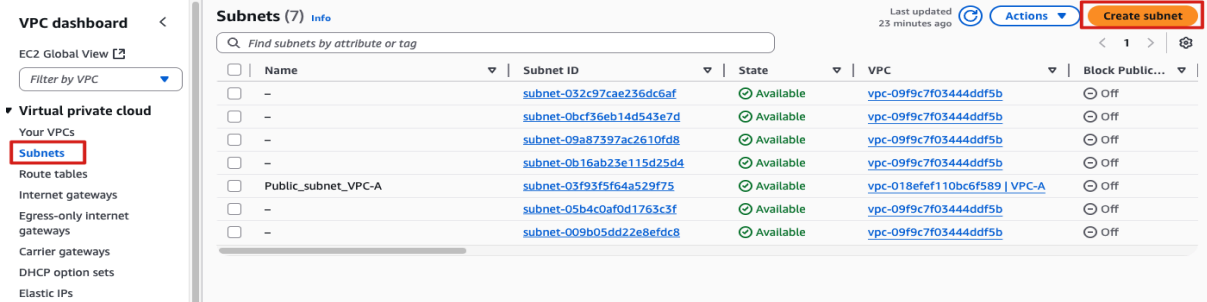
Cancel

Save

Step 8: VPC-B içinde Private Subnet Oluşturma

Bu subnet internete çıkmayacak, yalnızca iç iletişim için kullanılacak.

1. Subnets → Create Subnet seçin.



The screenshot shows the AWS VPC console. On the left sidebar, under 'Virtual private cloud', the 'Subnets' link is highlighted. The main area displays a table of subnets. The table has columns for Name, Subnet ID, State, VPC, and Block Public... The subnets listed include 'Public_subnet_VPC-A' and several private subnets. The 'Create subnet' button is highlighted in the top right corner.

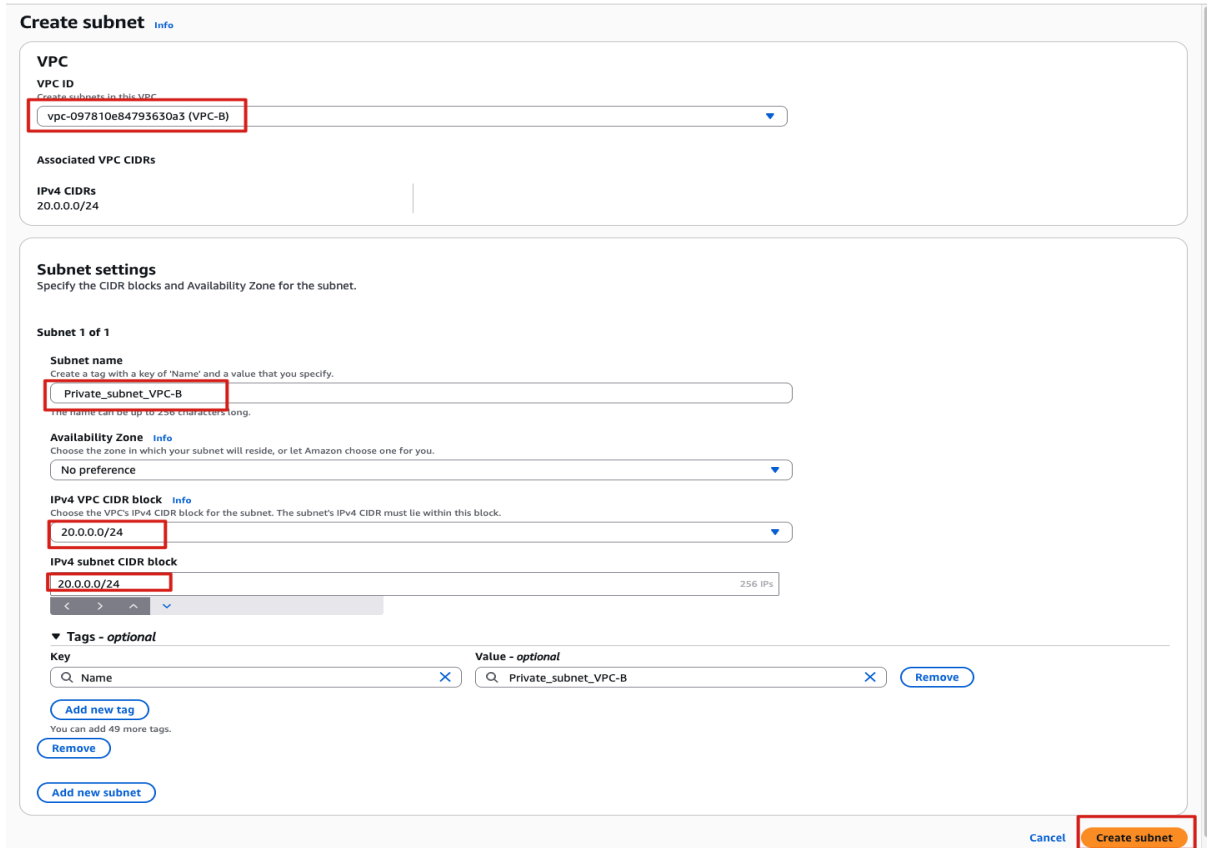
2. VPC ID: VPC-B

3. Subnet name: Private_subnet_VPC-B

4. Availability Zone: No Preference

5. IPv4 CIDR block: 20.0.0.0/24

6. Create subnet.



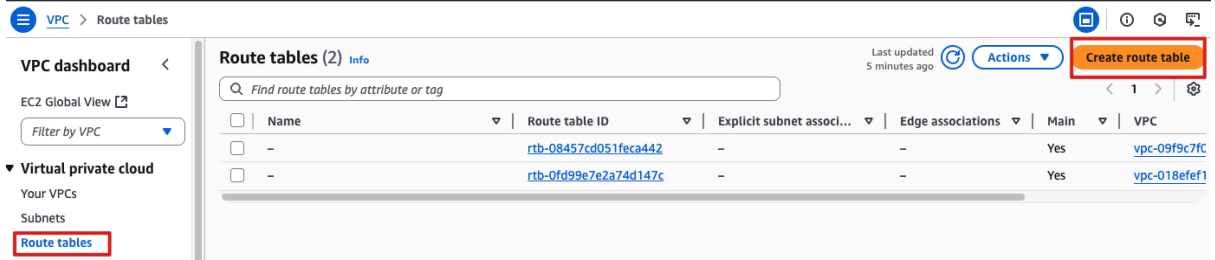
The screenshot shows the 'Create subnet' form in the AWS VPC console. The form is filled out with the following values:

- VPC ID:** vpc-097810e84793630a3 (VPC-B)
- Associated VPC CIDRs:** 20.0.0.0/24
- Subnet name:** Private_subnet_VPC-B
- Availability Zone:** No preference
- IPv4 VPC CIDR block:** 20.0.0.0/24
- IPv4 subnet CIDR block:** 20.0.0.0/24

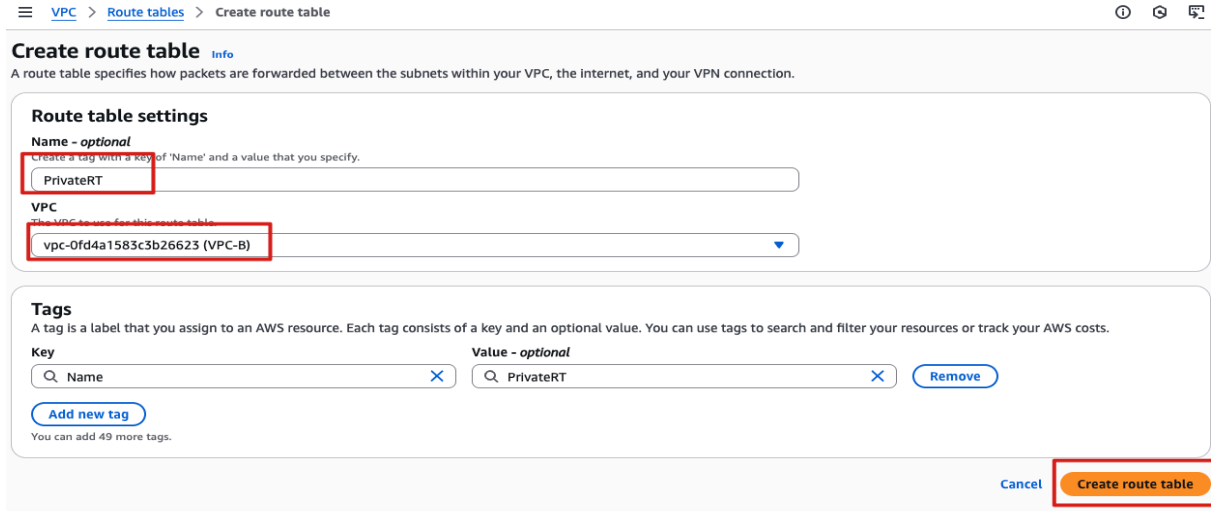
The 'Create subnet' button is highlighted in the bottom right corner.

Private Route Table Oluştur ve Subnet ile İlişkilendir

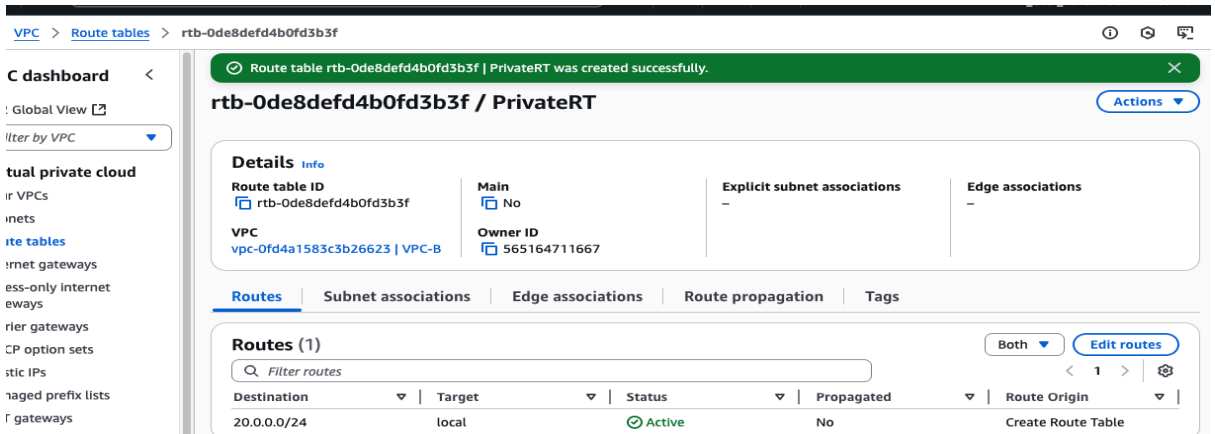
1. Sol panel: **Route tables** → **Create route table**



2. Name: **PrivateRT**
3. VPC: **VPC-B** → **Create route table**.



4. Alttaki **Subnet associations** sekmesine geç.
5. **Edit subnet associations** → **Private_subnet_VPC-B** seç → **Save associations**.



VPC > Route tables > rtb-0de8defd4b0fd3b3f > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Private_subnet_VPC-B	subnet-04c72ecff86d87829	20.0.0.0/25	-	Main (rtb-0594bfccb4ef3bedb)

Selected subnets

subnet-04c72ecff86d87829 / Private_subnet_VPC-B X

Cancel Save associations

Step 9: VPC-B Private Subnet'te EC2 Başlat

Şimdi VPC-B'nin private subnet'i içinde bir test sunucusu açalım.

1. Launch instances.

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch Instances

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	VPC-A_EC2	i-069dec671266f688e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d

2. Name: VPC-B_EC2

3. AMI: Amazon Linux 2023 (kernel-6.1)

Name and tags Info

Name

VPC-B_EC2

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

aws

Mac

ubuntu

Microsoft

Red Hat

SUSE

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-00ca32bbc84273381 (64-bit (x86), uefi-preferred) / ami-0aa7db6294d00216f (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250818.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-00ca32bbc84273381	2025-08-13	ec2-user

4. Instance type: **t2.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

5. Key Pair: listeden **ec2_ssh_key**'i seç.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ec2_ssh_key

[Create new key pair](#)

6. Network settings → Edit:

- **VPC:** **VPC-B**
- **Subnet:** varsayılan (Private_subnet_VPC-B)
- **Auto-assign public IP:** **Disable**
- **Firewall (SG):** **Create a new security group**
 - **Name:** **Private_EC2_SG**
 - **Description:** **Security group for private EC2**
 - **Rule:** **SSH** → **Source:** Anywhere (lab için)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-097810e84793630a3 (VPC-B)
20.0.0.0/24

[Create new VPC](#)

Subnet [Info](#)

subnet-06ad38a0e772d27ac Private_subnet_VPC-B
VPC: vpc-097810e84793630a3 Owner: 834266228480
Availability Zone: us-east-1b (use1-az6) Zone type: Availability Zone
IP addresses available: 251 CIDR: 20.0.0.0/24

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

Private_EC2_SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@!+=&:[]\$*

Description - required [Info](#)

Security group for private EC

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

7. **Launch instance** → **running** olana kadar bekle.

Instances (2) Info		Last updated 4 minutes ago Refresh	Connect	Instance state ▼	All states
☐	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	
☐	VPC-B_EC2	i-0c07615d88be38571	Running 🔍 🔍	t2.micro	
☐	VPC-A_EC2	i-0ec2dc90085c6e81	Running 🔍 🔍	t2.micro	

8. Bu makine **yalnızca Private IP** alır. **IPv4 Private IP**'yi not et

Instance summary for [i-0c07615d88be38571 \(VPC-B_EC2\)](#) [Info](#)

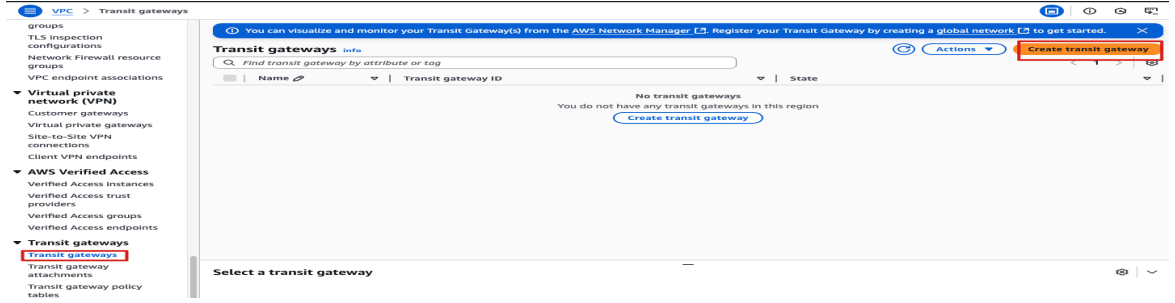
[Refresh](#) [Connect](#) [Instance state](#) ▼ [Actions](#) ▼

Updated less than a minute ago

Instance ID i-0c07615d88be38571	Public IPv4 address -	Private IPv4 addresses 20.0.0.20
IPv6 address -	Instance state 🕒 Pending	Public DNS -
Hostname type IP name: ip-20-0-0-20.ec2.internal	Private IP DNS name (IPv4 only) ip-20-0-0-20.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding 🔍 Opt-in to AWS Compute Optimizer for recommendations. Learn more 🔗
Auto-assigned IP address -	VPC ID vpc-0fd4a1583c3b26623 (VPC-B) 🔗	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-04c72ecff86d87829 (Private_subnet_VPC-B) 🔗	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:565164711667:instance/i-0c07615d88be38571	
Operator		

Step 10: Transit Gateway Oluřturma

1. **Services** → **VPC** → sol menüde **Transit gateways**.
2. **Create Transit gateway**.



- **Name tag:** DemoTG
- **Description:** TG for peering two VPCs
- Diğerlerini varsayılan bırak → **Create transit gateway**.

Create transit gateway [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details - optional

Name tag
Creates a tag with the key set to Name and the value set to the specified string.

DemoTG

Description [Info](#)
Set the description of your transit gateway to help you identify it in the future.

TG for peering two VPCs

Configure the transit gateway

Amazon side Autonomous System Number (ASN) [Info](#)

ASN

☒ **DNS support** [Info](#)

☐ **Security Group Referencing support** [Info](#)

☒ **VPN ECMP support** [Info](#)

☒ **Default route table association** [Info](#)

☒ **Default route table propagation** [Info](#)

☐ **Multicast support** [Info](#)

Configure cross-account sharing options

☐ **Auto accept shared attachments** [Info](#)

Transit gateway CIDR blocks

CIDR - optional [Info](#)

10.0.0.0/24

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

DemoTG

Add new tag

You can add up to 49 more tags.

Cancel **Create transit gateway**

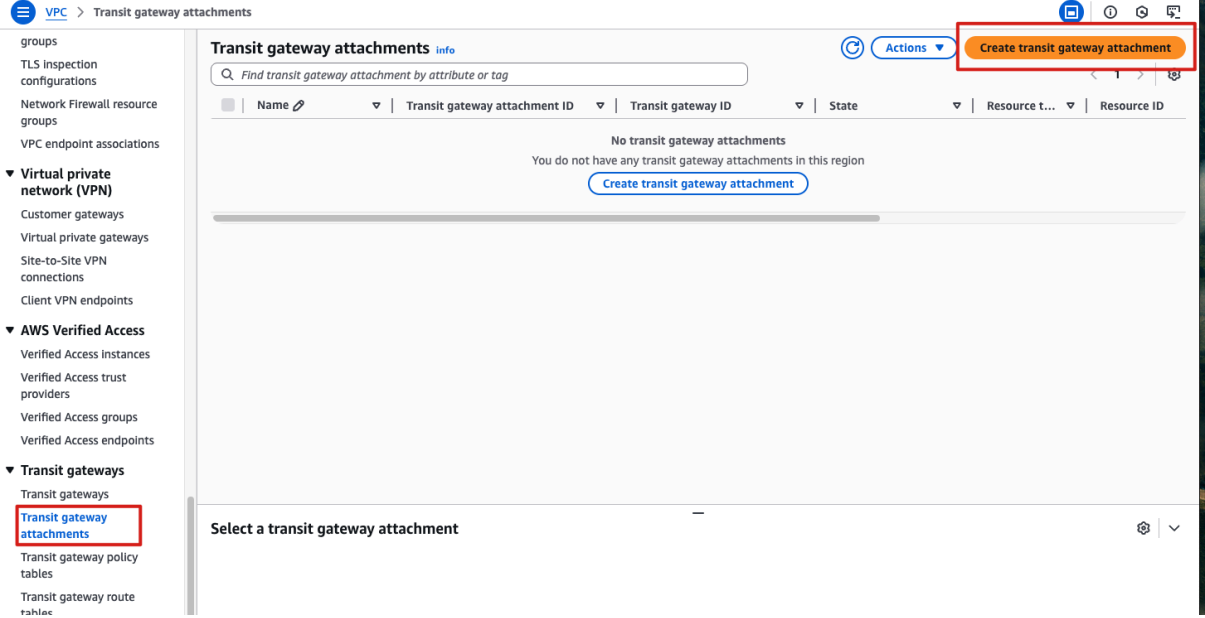
3. Durum **pending** → **available** olana dek birkaç dakika sürebilir.

Transit gateways (1) Info			Actions	Create
Find transit gateway by attribute or tag				
☐	Name	Transit gateway ID	State	
☐	DemoTG	tgw-01087849fdab85bb3	Available	

Step 11: VPC-A ve VPC-B için 2 TGW Attachment Oluşturma

Transit Gateway'in hem VPC-A hem de VPC-B'ye bağlanması için attachment oluşturmalıyız.

1. Sol menü: **Transit gateway attachments** → **Create transit gateway attachment**



- **Name tag:** VPC-A_TGA
- **Transit Gateway ID:** DemoTG
- **Attachment type:** VPC
- **DNS support:** Checked
- **IPv6 support:** Unchecked
- **VPC ID:** VPC-A
- **Subnet IDs:** Default

☰ VPC > Transit gateway attachments > Create transit gateway attachment

🔍 ⌂ 📄

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

VPC-A_TGA

Transit gateway ID [Info](#)
tgw-01087849fdab85bb3

Attachment type [Info](#)
VPC

VPC attachment

Select and configure your VPC attachment.

☒ **DNS support** [Info](#)

☒ **Security Group Referencing support** [Info](#)

☐ **IPv6 support** [Info](#)

☐ **Appliance Mode support** [Info](#)

VPC ID
Select the VPC to attach to the transit gateway.
vpc-0aa0ac3399337e18f

Subnet IDs [Info](#)
Select the subnets in which to create the transit gateway VPC attachment.

☐ us-east-1a	No subnet available
☐ us-east-1b	No subnet available
☐ us-east-1c	No subnet available
☐ us-east-1d	No subnet available
☐ us-east-1e	No subnet available
☒ us-east-1f	subnet-0190f36d0b573add9

subnet-0190f36d0b573add9

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional
Q Name	Q VPC-A_TGA

[Add new tag](#) [Remove](#)

You can add up to 49 more tags.

[Cancel](#) [Create transit gateway attachment](#)

Create transit gateway attachment butonuna basın

- **available** olması ~ birkaç dk sürebilir.

2. İkinci attachment: **Create transit gateway attachment**

- **Name tag:** VPC-B_TGA
- **Transit Gateway ID:** DemoTG
- **Attachment type:** VPC
- **DNS support:** Checked
- **IPv6 support:** Unchecked
- **VPC ID:** VPC-B
- **Subnet IDs:** Default

VPC > Transit gateway attachments > Create transit gateway attachment

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

VPC-B_TGA

Transit gateway ID

Info

tgw-03f23c94f235a6322

Attachment type

Info

VPC

VPC attachment

Select and configure your VPC attachment.

☒ DNS support

Info

☒ Security Group Referencing support

Info

☐ IPv6 support

Info

☐ Appliance Mode support

Info

VPC ID

Select the VPC to attach to the transit gateway.

vpc-097810e84793630a3

Subnet IDs

Info

Select the subnets in which to create the transit gateway VPC attachment.

☐ us-east-1a

No subnet available

☒ us-east-1b

subnet-06ad38a0e772d27ac

☐ us-east-1c

No subnet available

☐ us-east-1d

No subnet available

☐ us-east-1e

No subnet available

☐ us-east-1f

No subnet available

subnet-06ad38a0e772d27ac

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

X

Q VPC-B_TGA

X

Remove

Add new tag

You can add up to 49 more tags.

Cancel

Create transit gateway attachment

Create transit gateway attachment butonuna basın

- available olması ~ birkaç dk sürebilir.

Transit gateway attachments (2) info

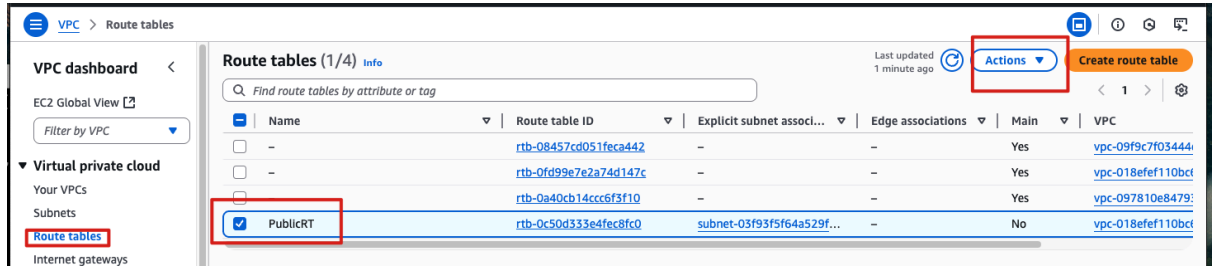
Find transit gateway attachment by attribute or tag

☐	Name	Transit gateway attachment ID	Transit gateway ID	State
☐	VPC-B_TGA	tgw-attach-0142079891700bb67	tgw-01087849fdab85bb3	Available
☐	VPC-A_TGA	tgw-attach-062e7b9669fe8192f	tgw-01087849fdab85bb3	Available

Step 12: VPC-A Route Table'a (PublicRT) VPC-B Rotasını Ekle

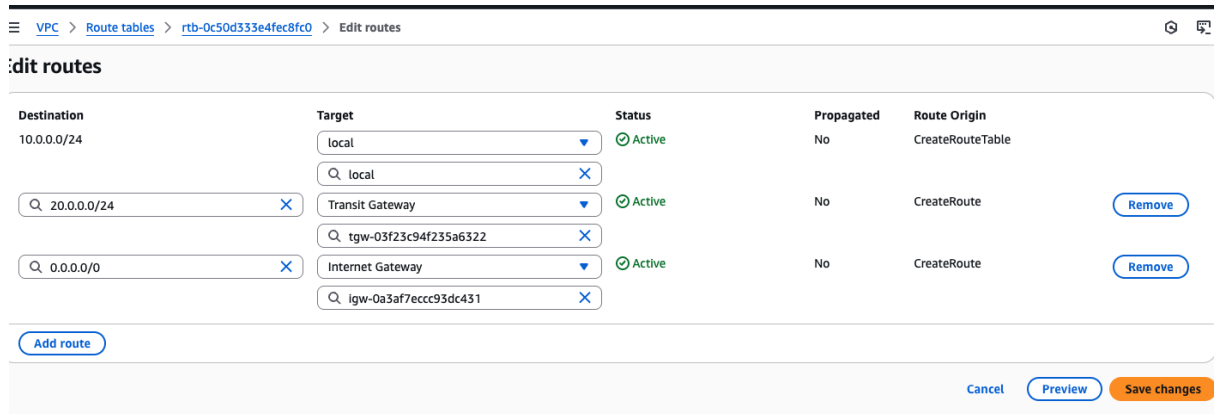
Amaç: VPC-B'nin CIDR'ine giden trafiği TGW'ye yönlendirmek.

1. **Route tables** → karışıklığı önlemek için **VPC filtresi** ile **VPC-A**'nın tablosunu süz.
2. **PublicRT** (subnet association'ı olan custom/non-default tablo) → **Route table ID**'ye tıkla.



3. **Routes** → **Edit routes** → **Add route**
 - **Destination:** 20.0.0.0/24 (VPC-B CIDR)
 - **Target:** Transit Gateway (DemoTG)
4. **Save changes.**
5. **Routes**'ta 20.0.0.0/24 girişi göründüğünü doğrula.

🔍 **Not:** Bazen hedef/destination anında görünmeyebilir; tekrar **Add route** işlemini deneyin.



Step 13: VPC-B Route Table'a VPC-A Rotasını Ekle

Amaç: VPC-A'nın CIDR'ine giden trafiği TGW'ye yönlendirmek.

1. **Route tables** → **VPC filtresi** ile **VPC-B**'yi seç.
2. VPC-B'de ekstra route table yoksa **main (default) route table** tek başınadır.
3. **Route table ID** → **Routes** → **Edit routes** → **Add route**

Destination: 20.0.0.0/24

Target: local

Status: Active

Propagated: No

Route Origin: CreateRouteTable

Destination: 10.0.0.0/24

Target: Transit Gateway

Status: Active

Propagated: No

Route Origin: CreateRoute

Remove

Add route

Cancel Preview Save changes

- **Destination:** 10.0.0.0/24 (VPC-A CIDR)
 - **Target:** Transit Gateway (DemoTG)
4. **Save changes.**
 5. **Routes**'ta 10.0.0.0/24 girişi görüldüğünü doğrula.

rtb-0de8defd4b0fd3b3f / PrivateRT

Actions

Details

Route table ID: rtb-0de8defd4b0fd3b3f

VPC: vpc-0fd4a1583c3b26623 | VPC-B

Main: No

Owner ID: 565164711667

Explicit subnet associations: -

Edge associations: -

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Both

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	tgw-01087849fdab85bb3	Active	No	Create Route
20.0.0.0/24	local	Active	No	Create Route Table

✓ Böylece iki VPC Transit Gateway'ye bağlanmış oldu.

Step 14: İki VPC Arası Bağlantıyı Test Et

Bu adım, Transit Gateway ile VPC-A ve VPC-B'nin başarıyla konuşurulduğunu doğrular.

Bilgisayarınızdan VPC-A_EC2'ye bağlanmak isterseniz önce key dosyasının izinlerini ayarlamalısınız.

```
chmod 400 ec2_ssh_key.pem
```

Local makinenizden private key dosyasını bu sunucuya kopyalayın:

```
scp -i ec2_ssh_key.pem ec2_ssh_key.pem  
ec2-user@ec2-3-236-83-74.compute-1.amazonaws.com:/home/ec2-user/
```

Şimdi bu sunucuya bağlanalım

```
ssh -i "ec2_ssh_key.pem"  
ec2-user@ec2-3-236-83-74.compute-1.amazonaws.com
```

```
(base) Downloads ➤ ssh -i "ec2_ssh_key.pem" ec2-user@ec2-3-236-83-74.compute-1.  
amazonaws.com  
  
A newer release of "Amazon Linux" is available.  
Version 2023.8.20250908:  
Run "/usr/bin/dnf check-release-update" for full release and version update info  
  
#_  
~\_ #####_ Amazon Linux 2023  
~~ \_#####\  
~~ \_###|  
~~ \#/ ____ https://aws.amazon.com/linux/amazon-linux-2023  
~~ V~' '->  
~~~ /  
~~._. _/  
_/_/_/  
_/_/m/'  
  
Last login: Wed Sep 10 11:44:54 2025 from 31.155.252.140  
[ec2-user@ip-10-0-0-15 ~]$  
[ec2-user@ip-10-0-0-15 ~]$ ls -l  
total 4  
-r----- . 1 ec2-user ec2-user 1674 Sep 10 11:46 ec2_ssh_key.pem  
[ec2-user@ip-10-0-0-15 ~]$
```

Sunucuya bağlanınca dosyanın izinlerini sıkılaştırın:

```
chmod 400 ec2_ssh_key.pem
```

